

STRATA implementation audit

Module-by-module verification of the STRATA strategy execution platform against the Executive Product Blueprint and Enterprise Data Lineage Model. Evidence gathered from the live application (localhost:8080/strata), the 9 STRATA migration files, the domain/type layer, and the feature folder's own execution log.

Method live-app walkthrough (10 screens + 9 admin tabs) · direct migration reads · type-layer inspection

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Scope code + schema; no production DB inspected (staging-only per handover)

Verdict: substantially blueprint-compliant, not a demo shell

- The architecture is config-first and evidence-linked in practice, not just in intent — verified in running code, not claimed. Real gaps remain in identity resolution, tenancy, and two backend services that the UI already has surface area for. Details below.

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1 · EXECUTIVE SUMMARY

What is actually built

STRATA's data layer is a real, config-first domain model: 54 `strata_*` tables covering strategy, scorecards, KPI/OKR, execution, portfolio/value, lineage, and governance — matching the blueprint's entity catalogue (§15–17) almost one-for-one. Every governed record (perspectives, scorecard models, thresholds, value categories, gate models, KPI types, upload templates, workflows) carries a `GovernedEnvelope`: version, status, effective dates, approver, change reason, supersession pointer. Separation of duties is enforced with database `CHECK` constraints, not UI convention — e.g. `validator_id <> owner_id` on `benefits`, `validated_by <> submitted_by` on

benefit values and KPI actuals. The calculation engine's own header comment states the non-negotiable directly: *"UI NEVER computes enterprise scores... no RAG constants anywhere in this engine."* That was verified against the running Command Center, Scorecards, and KPI Library — every score, band, and rollup traces to a `strata_calculated_values` row or a frozen snapshot payload.

All 10 executive screens are live against real (seeded) data, not static mockups: the Strategy Room surfaces a `CHARTER INCOMPLETE` flag on plays lacking a charter — the exact governance rule blueprint §6 demands ("a strategic play without a charter... should not be promotable"). The Scorecard detail page carries a "Frozen in snapshot" banner and a dedicated `/evidence` route stating *"Every number traces to source, formula version, validation and snapshot."* The Data pipeline page renders the full 8-stage lineage stepper (Source → Ingestion → Staging → Validation → Attestation → Canonical write → Calculation → Snapshot) with a real upload run (RUN-1001, 8 valid / 1 rejected of 9 raw rows).

What is only visual / demo

The seed data is Salam-modeled but is consistently and visibly labeled `DEMO SEED` (Salam reference tenant) on every governed record in the Admin console — it does not leak into core code, route names, or component names (confirmed by a targeted grep: zero "Salam" references inside `src/modules/strata/`). Two capabilities have UI surface area but no live backend behind them yet: the AI advisory output shown on Command Center ("ADVISORY (draft — requires human review)... Confidence 72%") is a seeded row against a real, provenance-carrying schema (`strata_ai_outputs`) — there is no generation service producing new advisories. Similarly the Jira-sourced project card (NET5G, 95% mapping confidence) is seed data against a genuinely source-agnostic schema — there is no live Jira connector syncing it.

What is missing

Organization/tenant scoping is present as a nullable column on every table but not populated or enforced — a disclosed, deliberate deferral ("Single-tenant deployment (Q4): `organization_id` kept nullable for future use," per migration comment) rather than a hidden gap, but it does mean the "reusable across organizations" claim is not yet true at the data-isolation level. The audit trail has no actor/user column. Frozen evidence rows in Reviews & Decisions show entity *type* but not entity *name*. OKRs are fully modeled but entirely unpopulated (zero rows) — the feature is unproven end-to-end. No user currently holds a STRATA role in the running environment ("Your roles: —"), so the SoD model has not been exercised by a real multi-person workflow yet.

Blueprint-compliant: qualified yes

Architecturally compliant on every non-negotiable this audit could test directly (configuration-first, evidence-linked, snapshot-aware, SoD-enforced, zero hard-coded taxonomy). Not yet complete

on organizational reusability, identity resolution, and the two backend services (AI generation, live Jira sync) that the UI is already wired for.

Top 10 gaps, ranked

- 1 **P1** **No live AI advisory generation service.** The schema, provenance fields, and UI surface (Command Center "Needs attention") all exist; nothing currently produces a new `strata_ai_outputs` row from live data.
- 2 **P1** **No live Jira connector.** Project Card is genuinely source-agnostic in schema, but the one `source_system='jira'` card in the running app is seed data — sync is not live.
- 3 **P1** **Tenant isolation is schema-present but inactive.** `organization_id` is nullable and unenforced across all 54 tables — disclosed as deferred, but blocks the "reusable across organizations" claim today.
- 4 **P2** **No STRATA role is granted to any live user.** RBAC exists as a schema and an admin Roles tab, but "Your roles: —" in the running app means SoD has not been exercised end-to-end by real people.
- 5 **P2** **Audit trail has no actor column.** `strata_audit_events` records entity/action/timestamp but the Change log UI shows no "by whom" — undermines the governance trail's usefulness for a real review.
- 6 **P2** **Reviews frozen-evidence rows show entity type, not entity name.** A CEO reading a locked snapshot sees "Kpi — Achievement pct — 77.5" with no way to tell which KPI without leaving the page.
- 7 **P2** **OKR/KeyResult feature is unpopulated.** Fully modeled (interoperable with KPI per blueprint §8) but zero seeded rows — creation flow and rollup behavior are unverified.
- 8 **P2** **Multi-cycle behavior unverified.** Only one `StrategyCycle` (FY2026) exists; "clone prior cycle" and multi-year strategy support (blueprint §6) could not be exercised.
- 9 **P3** **Ownership fields render as "—" for at least one KPI.** Likely correct zero-assumption behavior for an unassigned field rather than a resolution bug, but was not distinguishable from a broken lookup during this audit — needs a populated-owner KPI to confirm.
- 10 **P3** **Known, disclosed regression already fixed at source** (ads DropdownMenu wrapper dropping `triggerRef`) — worked around via a local component, flagged separately for the design-system owner. Listed here only for completeness; not a STRATA defect.

Coverage against the blueprint's five architectural layers

Legend

built & verified

partial / schema-only

not built

MODULE	SUBMODULE	BUILT?	REAL OR MOCK DATA?	CONFIG-DRIVEN?	LINE
Executive Experience	Command Center	yes	Seeded, calc-engine values	yes	yes (links
	Strategy Room / Map	yes	Seeded	yes	via s elem
	Board Pack export	yes	Generated client-side	tied to snapshot_id	yes
Strategy & Scorecard Domain	Perspectives / Scorecard models	yes	DEMO SEED, versioned	yes	n/a
	KPI / OKR Library	partial	KPI real; OKR empty	yes	yes (
Execution & Value Domain	Initiative Registry / Project Cards	yes	Seeded, source-agnostic schema	partial (stage is workflow-config-driven per comment)	map field
	Portfolio / VMO / Value Gates	yes	Seeded	yes (gate models, value categories)	yes
Governance / Admin / AI	Admin Console (9 config areas)	yes	All DEMO SEED, versioned/approved	yes — this is the config layer	n/a
	Review Cadence / Snapshots / Decisions	yes	SNAP-1001 real locked snapshot	n/a	30 f
	AI Insights	schema + UI only	1 seeded advisory row	n/a	entit snap conf exist
Integration / Data Foundation	Uploads / Staging / Validation	yes	RUN-1001 real run, row-level errors	yes (upload templates)	yes -

MODULE	SUBMODULE	BUILT?	REAL OR MOCK DATA?	CONFIG-DRIVEN?	LINE
	Connectors (Jira/API/ERP)	not live	seed only	n/a	sync exist
	Formula / Calculation Engine	yes	Real, server-side only	yes — reads threshold_scheme, model records	full p (form input sour

3 · HARD-CODE & DEMO DATA AUDIT

Constants and seed data checked against the configuration-first rule

A targeted search for hard-coded taxonomy arrays (perspective names, RAG bands, scorecard model names, gate stages) inside `src/modules/strata/` returned **zero matches**. The `types` file states the rule at its own header: *"Business truth (band keys, stage names, perspective names, weights, formulas) is NEVER encoded here — those arrive as governed config records at runtime."* The unions that do exist in `types.ts` (e.g. `'draft' | 'pending_approval' | 'approved' | 'retired' | 'superseded'`) mirror database CHECK constraints on system-level workflow states, not tenant-specific business taxonomy — this is the correct place to hard-code, since these are protocol states, not configurable content.

LOCATION	WHAT'S THERE	IS IT A VIOLAT
<code>types.ts</code>	GovernedStatus, DataState, ValidationStatus unions	no
<code>strata_role_assignments</code> (migration)	6-value CHECK on role (<code>strata_admin</code> , <code>strategy_office</code> , <code>executive_viewer</code> , <code>kpi_owner</code> , <code>vmo_validator</code> , <code>data_steward</code>)	borderline
<code>strata_project_cards.source_system</code> CHECK	<code>'jira', 'manual', 'upload', 'api'</code>	no

LOCATION	WHAT'S THERE	IS IT A VIOLATION
strata_benefits.lifecycle_stage CHECK	9-stage fixed enum	no
20260705100600_strata_seed_salam_demo.sql	Full demo dataset: perspectives, scorecard model, 8 KPIs, 3 initiatives, 4 project cards, portfolio + 3 benefits, 2 gates, 1 locked snapshot	no — labeled
strata_threshold_schemes seed row	Named "Salam Standard RAG"	cosmetic
src/modules/strata/ (full grep)	Searched for "Salam" string	zero matches

Verified via direct file reads of all 9 migrations and types.ts, plus a repo-wide grep for perspective-name literals and "Salam" inside the strata module. Not exhaustively re-verified against every page component's JSX for one-off demo strings (e.g. placeholder copy) — a full lint-style sweep of the 14 page files was not completed in this pass.

4 · DATA MODEL AUDIT

Blueprint entity catalogue vs. the 54 strata_* tables

REQUIRED ENTITY	EXISTS?	TABLE	NOTES /
Tenant / Organization	nullable column only	organization_id on every table	Not a standard enforced Document

REQUIRED ENTITY	EXISTS?	TABLE	NOTES /
			"Single-f (Q4)"
StrategyCycle	yes	strata_cycles	Only one (FY2026 path unv
Period	yes	strata_periods	—
Perspective	yes	strata_perspectives	Full Gove
StrategyElement	yes	strata_strategy_elements	Generic (Theme/ observed
StrategyMapEdge	yes	strata_map_edges	Confider animatio
ScorecardModel / Instance / Line	yes	strata_scorecard_models / _instances / _lines	Model-p validated approval
KPI / FormulaVersion / Target / Actual	yes	strata_kpis / _kpi_formula_versions / _kpi_targets / _kpi_actuals	5 distinc KPI; form approval
Objective	subtype, not standalone	strata_strategy_elements (typed)	Modeled Strategy matching "generic pattern"
OKR / KeyResult	schema yes, data no	strata_okrs / strata_key_results	Zero row OKRs ye
Commentary	yes	strata_commentary	Not direc walkthro
Initiative	yes	strata_initiatives	value_hy present
ProjectCard	yes	strata_project_cards	Source-e UNIQUE source_k "STRATA connecto
Milestone	yes	strata_milestones	Weighte
Dependency	yes	strata_dependencies	SLA days
ExecutionLink	yes	strata_execution_links (+ *_elements/*_kpis/*_projects link tables)	Generic junction

REQUIRED ENTITY	EXISTS?	TABLE	NOTES /
Portfolio / PortfolioMembership	yes	strata_portfolios / _portfolio_memberships	—
Benefit / Baseline / Target / Actual	yes	strata_benefits / strata_benefit_values (value_kind enum unifies baseline/planned/forecast/realized)	9-stage CHECK
ValueHypothesis / CausalMechanism	fields, not tables	strata_benefits.value_hypothesis / causal_mechanism	Text field rather than entity —
Assumption	yes	strata_assumptions	Confidence (open/hypothesis)
AttributionRule	yes	strata_attribution_rules	rule_type counterfactual double-blind
ValueGate	yes	strata_gate_models / _gate_model_stages / _gate_instances / _gate_evidence	5-stage model of approval
Snapshot	yes	strata_snapshots / _snapshot_items	INSERT- RPC; SNAPSHOT
Decision / Action	yes	strata_decisions / strata_actions	DEC-/ACTION
DataSource / UploadRun / ValidationResult	yes	strata_data_sources / _upload_runs / _validation_results	File hash status message
LineageRecord	yes	strata_lineage_records	configurable template active at time
ConfidenceScore	field, not entity	confidence numeric column on ~8 tables	Per-record scored equivalent
AuditEvent	yes, incomplete	strata_audit_events	Entity/audit present; column schema log UI

All 10 executive screens, walked live against staging data

Command Center

/strata

Observed	Enterprise score 96.1 (ring + "View evidence"), Value at risk SAR 2.7M (evidence-linked), Benefits realization 86.3%, Open decisions/Blocked dependencies/Pending attestations as live counts, Perspective health (5 rows, weight + score + band), Enterprise score trend (2-point line), Needs attention (Exception / Action / Advisory rows)
Gap	None found against blueprint §1 requirements. The AI advisory row is explicitly labeled "draft — requires human review" with a confidence score, matching §13's provenance requirement

Strategy Room + Strategy Map

/strata/strategy, /strata/strategy/map

Observed	Theme → Play → Objective hierarchy; CHARTER INCOMPLETE flag on plays lacking a charter; KPI coverage panel; Cause & effect table with confidence %; real xyflow canvas with perspective-colored nodes and Drives/Contributes-to/Enables edge types, confidence-gated animation
Gap	None found. This is the clearest live proof of blueprint §6's governance rule ("a play without charter/owner/KPI/value-hypothesis should not be promotable")

Scorecards (list + detail + evidence)

/strata/scorecards, /scorecards/:slug, /scorecards/:slug/evidence

Observed	LOCKED (87.5) vs LIVE (96.1) instances with distinct scores; "Frozen in snapshot" banner on locked instance; perspective breakdown + 8-line drilldown table w/ weight/performance/score/band; dedicated evidence route stating "nothing here is computed in the UI"
Gap	None found in the drilldown chain itself. Minor: evidence "Inputs" list rounds to 1 decimal consistently now (an earlier defect noted unrounded values — appears fixed)

KPI & OKR Library (+ detail)

/strata/kpis, /kpis/:slug

Observed	8 KPIs w/ unit, direction, frequency, entry method, status, achievement; formula version (v1, expression shown, "every formula change creates a new approved version"); lineage table (period, value, entry method, upload run, staging ref, validated-at)
Gap	P3 Ownership block (accountable/data/reporter/validator/escalation owner) shows "—" for all 5 roles on the KPI inspected. Could not confirm whether this is correct zero-assumption rendering of an unassigned field or a resolution bug — needs a KPI with an assigned owner to distinguish. P2 OKR section is empty (0 rows) — feature unproven

Execution

/strata/execution

- Observed 3 initiatives w/ budget/stage/strategy-linkage; project card "5G Rollout Wave 2 · Jira · NET5G" w/ mapping confidence 95%, progress bar, milestone count, "synced 1d ago"; dependencies panel
- Gap **P1** The Jira sync is not live — "synced 1d ago" is seed metadata, and the handover explicitly lists "Jira adapter" as a deferred backend slice

Portfolio & Value (VMO)

/strata/portfolio

- Observed Value target/at-risk/benefits/open-gates KPI row ("Server-calculated · View evidence" on Value at risk); benefit register (3, w/ category/lifecycle/confidence/realization); value profile table (baseline/planned/forecast/realized) with a PENDING validation state and the note "Realized values require independent finance validation"
- Gap **P3** Validator column shows "—" — same ownership-resolution ambiguity as the KPI detail page

Data & Lineage

/strata/data

- Observed Full 8-stage pipeline stepper; 2 registered sources (Salam BI Extract, Salam Finance Excel) w/ system type + refresh cadence + status; 1 upload run (RUN-1001, 8 valid / 1 rejected of 9 raw, COMPLETED)
- Gap None found against blueprint §17/§20. Only one upload run exists, so the "many runs over time" trend view is unverified but the mechanism is sound

Reviews & Decisions

/strata/reviews

- Observed Period governance banner ("Q2 FY2026 OPEN — closing is blocked while attestations are pending; overrides are audited"); SNAP-1001 locked snapshot w/ 30 frozen records (Kpi-16, Perspective-5, Scorecard instance-1, Scorecard line-8)
- Gap **P2** Frozen evidence rows show entity type + metric + value but not the entity's name — a reviewer can't tell which KPI scored 77.5 without cross-referencing elsewhere

Configuration / Admin

/strata/admin/*

(9 tabs)

- Observed Perspectives, Scorecard models ($\Sigma=100$ "WEIGHTS VALID"), Thresholds (RAG bands + tolerance + confidence threshold), Value taxonomy (5 categories w/ validator role + cadence), Gates (5-stage model w/ decision options + approver roles per stage), KPI types, Upload templates (column-mapping schema shown), Workflows (state-

machine w/ approval roles), Roles (6 personas, SoD note), Change log (20-event audit trail)

Gap

P2 Change log has no actor column. **P2** No visible "grant role to user" action was found in the Roles tab during this walkthrough — could not confirm an admin UI exists for assigning `strata_role_assignments`, only that the underlying table and 6-persona model do

Upload Wizard

/strata/data/upload

Observed

Per the execution log: 4-step wizard (xlsx/csv/paste), display-only pre-checks, writes to staging via RLS

Gap

P2 Per the same log, the server-side validation/promote RPC is stated as "next backend slice" at the time of that entry — a later migration (`20260705140000_strata_upload_validation_promote.sql`) exists and was not independently re-verified live in this audit

6 · FIX STRATEGY

Phased plan — architecture is sound; remaining work is service-completion and rollout, not a rebuild

Unlike a from-scratch remediation, this fix plan does **not** require a Phase 0 "stop the demo architecture" step — the architecture already is the governed, config-first model the blueprint specifies. The phases below focus on closing the specific gaps found above.

Phase 1 Identity & rollout

- Grant `strata_role_assignments` to at least the second-approver set so SoD flows (submit/approve, validate/attest) can be exercised by real distinct users, not just proven in schema
- Populate accountable/data/reporter/validator/escalation owners on at least one KPI and one benefit to distinguish "unassigned, correctly blank" from "resolution broken" (resolves the P3 ownership-rendering ambiguity)
- Add an actor column (or join) to the Change log UI — the underlying `strata_audit_events` data model likely already has a `created_by/actor` reference; confirm and surface it

Phase 2 Tenant isolation activation

- Decide whether multi-tenant is in scope for the next release slice; if yes, populate `organization_id` and add RLS predicates scoped to it across all 54 tables (the column already

exists everywhere — this is enforcement work, not schema work)

- If multi-tenant is out of scope for now, update the blueprint's "reusable across organizations" claim to reflect single-tenant-per-deployment as the current release boundary, so the gap is a documented scope decision rather than a silent shortfall

Phase 3 Backend services the UI is already wired for

- Build the AI advisory generation service against the existing `strata_ai_outputs` schema (`entity_refs`, `snapshot_id`, `confidence`, `human_review_status` are already correct) — the UI does not need to change, only a real producer of rows
- Build the live Jira connector against the existing source-agnostic `strata_project_cards/strata_execution_links` schema — again, schema and UI are ready; this is integration work
- Verify the upload validation/promote RPC (migration 20260705140000) end-to-end live, since the execution log's own note about it predates that migration

Phase 4 Evidence completeness

- Reviews & Decisions: resolve `entity_id` → entity name on frozen evidence rows (join against the relevant table at snapshot-read time, or denormalize the name into `strata_snapshot_items` at freeze time)
- Exercise the OKR/KeyResult creation flow with real data — confirm rollups and interoperability with KPI-linked key results actually work, not just that the schema supports it
- Exercise a second `StrategyCycle` (e.g. FY2027) to confirm the clone-prior-cycle and multi-year path blueprint §6 describes

Phase 5 QA & hygiene

- Physical deletion of dormant legacy StrategyHub/Astryx files and `es_*` tables (already logged as deferred cleanup, DRIFT-003)
- Migrate `StrataChipMenu` back to the ADS `DropdownMenu` wrapper once the design-system owner's fix for the dropped `triggerRef` lands
- Apply the STRATA migrations to production (currently staging-only per the feature's own handover) — an owner-approved step, not an engineering one

P0 — architecture violation

P1 — executive trust / lineage violation

P2 — major functional gap

P3 — UI/UX or usability gap

P4 — polish

SEVERITY	COUNT	ITEMS
P0	0	None found. No hard-coded taxonomy, no client-side scoring, no silent formula/threshold changes were located anywhere in this audit.
P1	3	No live AI advisory generation service · No live Jira connector · Tenant isolation schema-present but inactive/unenforced
P2	5	No STRATA role granted to any live user · Audit trail has no actor column · Frozen-evidence rows show type not name · OKR/KeyResult unpopulated · Multi-cycle path unverified
P3	2	Ownership-field rendering ambiguity (unassigned vs. broken resolution) · Known ads DropdownMenu wrapper bug (already worked around, owner notified separately)
P4	1	Dormant legacy file / es_* table cleanup (already logged as deferred, DRIFT-003)

8 · APPROVAL GATE

This audit found a governed, evidence-linked, configuration-first implementation that matches the blueprint's non-negotiables in the code that is actually running — not a hard-coded demo wearing a governance skin. The gaps are real but narrow: two backend services the UI already anticipates, tenant-isolation enforcement, identity/rollout exercises, and evidence-display polish. None of them require touching the domain model or reopening the architecture.

I will not implement fixes until you approve the fix strategy in section 6.